



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of Methodologies:** Data was collected from **SpaceX's API** and **web scraping**, processed via Python (requests, BeautifulSoup, Pandas) to extract launch details (e.g., booster version, payload mass, landing outcomes). **Data wrangling** handled missing values and encoded categorical variables. **Exploratory Data Analysis (EDA)** used SQL and visualizations (bar, pie, scatter plots) via Folium and Plotly Dash. **Predictive analysis** employed logistic regression, optimized with GridSearchCV, and evaluated with accuracy, precision, and ROC-AUC metrics.
- **Summary of Results:** EDA revealed Cape Canaveral (CCAFS SLC-40) as the most reliable launch site (42.9% success rate), with 98 successful missions and 1 failure overall. Payload mass showed no strong correlation with launch success. Early drone ship landings (2015) struggled, while ground pad landings succeeded by December 2015. Logistic regression achieved the highest classification accuracy (83.3%) for predicting launch success. Interactive Folium maps and Plotly Dash dashboards visualized launch site geography, success rates, and payload-outcome relationships.

Introduction

- **Project Background and Context:** This project analyzes SpaceX's Falcon 9 launch data to understand factors influencing mission and landing success, leveraging data science to explore reusable rocket technology's impact on cost-efficiency and reliability in the space industry.
- **Problems to Address:**
 - Which launch sites yield the highest **success rates**?
 - How do payload mass, orbit type, and booster version **affect mission** and landing outcomes?
 - Can we **predict launch success** using classification models?
 - What are the geographic and **infrastructural characteristics** of SpaceX's launch sites?

Section 1

Methodology

Methodology

Executive Summary

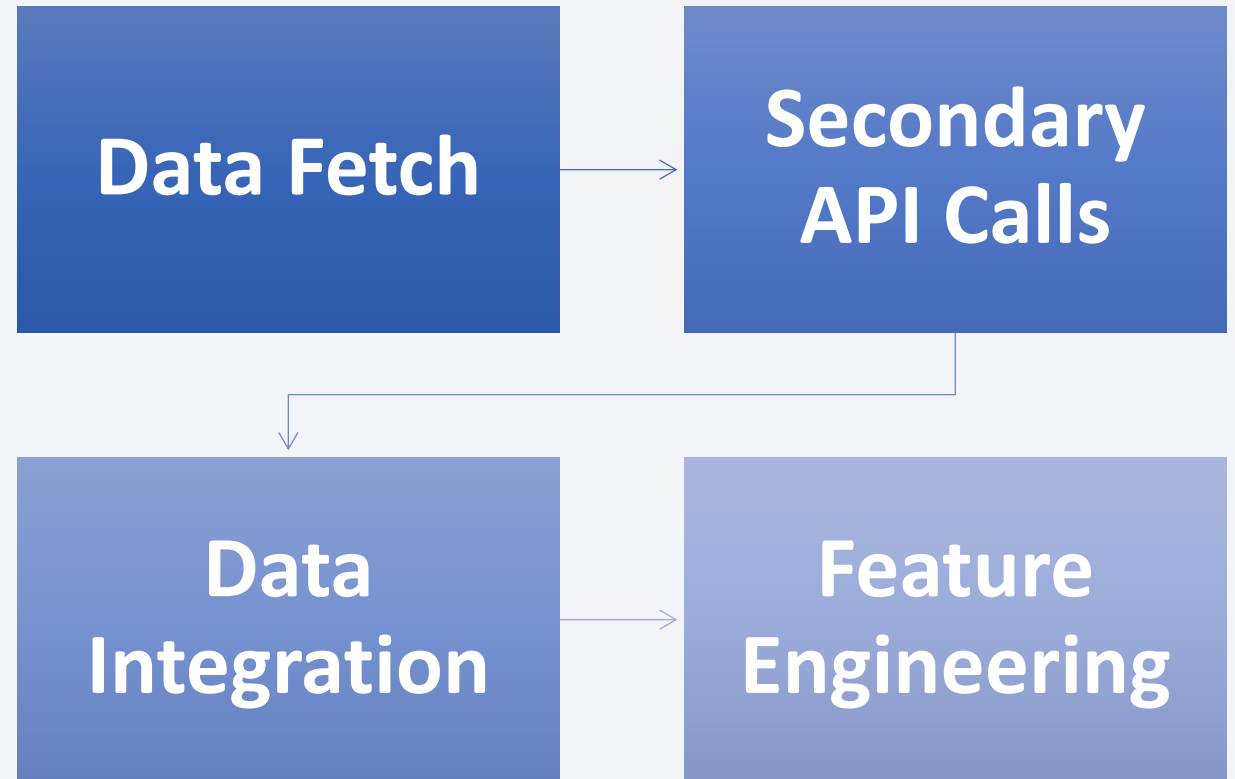
- Data collection methodology:
 - Data was collected via SpaceX's public API using Python's requests library, extracting core details (e.g., block, reuse count, serial) for each rocket launch
- Perform data wrangling
 - Handled missing values (e.g., None for missing cores), Structured data into lists (e.g., Block, ReusedCount) for analysis, Encoded categorical variables (e.g., landing outcomes) for modeling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Models: Logistic regression predicted launch success, Tuning: GridSearchCV optimized hyperparameters (e.g., regularization), Evaluation: Scikit-learn metrics (accuracy, precision, recall, ROC-AUC) assessed model performance.

Data Collection

- Data was collected from SpaceX's API (``api.spacexdata.com/v4/launches/past``) using Python's ``requests`` library. The raw JSON response was converted into a DataFrame, but key details (e.g., rocket name, payload mass) were only stored as IDs. Secondary API calls enriched this data by fetching:
 - - Rocket ID → Booster name (e.g., "Falcon 9")
 - - Payload ID → Mass (kg) and orbit
 - - Launchpad ID → Site name and coordinates
 - - Cores ID → Landing outcome, reuse count, and technical specs
- The final dataset was filtered for Falcon 9 launches, and missing payload masses were replaced with the mean value.

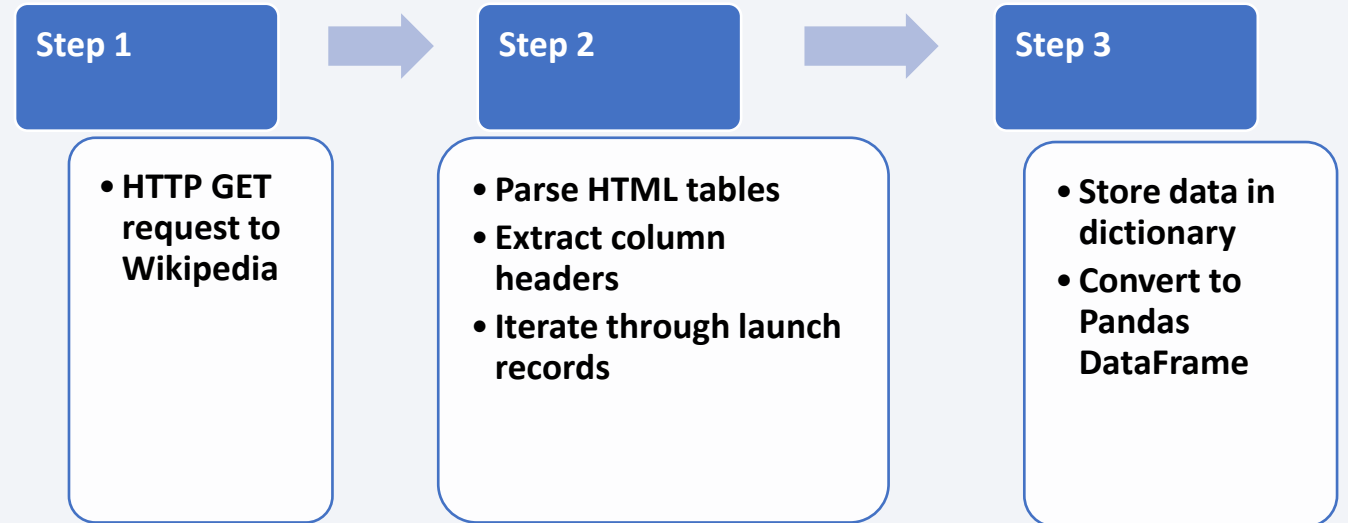
Data Collection – SpaceX API

- **API Endpoint Access:** Making GET requests to SpaceX's REST API endpoints
- **Data Extraction:** Retrieving launch, rocket, payload, and core data
- **Data Transformation:** Converting JSON responses into structured data
- **Feature Engineering:** Creating derived metrics from raw API data
- **Data Aggregation:** Combining multiple API responses into a unified dataset
- <https://github.com/FerAbrCode/Capstone/blob/main/DataCollection.ipynb>



Data Collection - Scraping

- The webscraping is done using **requests** and **BeautifulSoup**.
- Flight number, date/time, booster version, payload details (name, mass, orbit, customer), and launch/landing outcomes is extracted.
- Custom functions handle **data cleaning** and **unit conversions**.
- The processed data is **stored** in a Pandas **DataFrame** and exported to a **CSV** file for analysis.



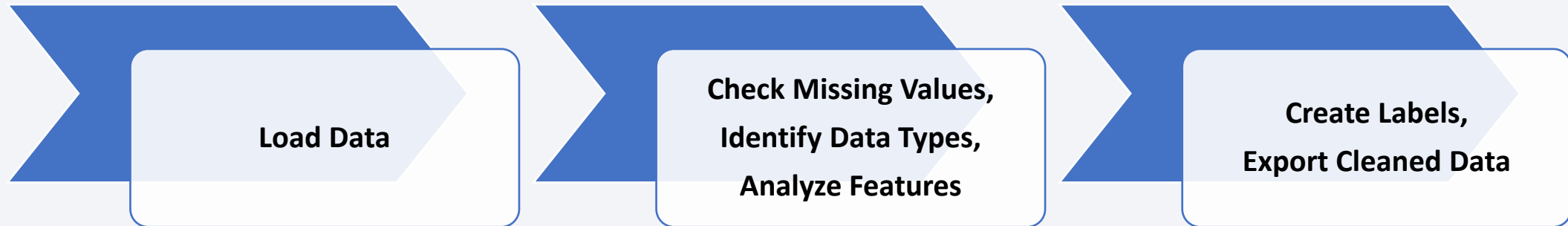
<https://github.com/FerAbrCode/Capstone/blob/main/Webscraping.ipynb>

Data Wrangling

The SpaceX Falcon 9 dataset was processed to **analyze** first-stage landing outcomes.

Steps included loading the CSV, **handling** 28.9% missing LandingPad values, **identifying** numerical (e.g., PayloadMass) and categorical (e.g., Orbit) columns, **analyzing features** with `value_counts()`, **creating binary** landing outcome labels (1=success, 0=failure), and **exporting** the cleaned **dataset** with the new Class label.

<https://github.com/FerAbrCode/Capstone/blob/main/DataWrangling.ipynb>



EDA with Data Visualization

Flight Number vs. Launch Site: Shows how launch site usage evolved over time

Payload vs. Launch Site: Compares payload mass distributions across sites

Success Rate vs. Orbit Type: Shows how early GTO, VLEO, and PO missions had lower success rates due to complexity and experimental nature

Flight Number vs. Orbit Type: Visualizes how orbit types shifted over Falcon 9's evolution

Payload vs. Orbit Type: Explores payload mass variability across orbit types

Launch Success Yearly Trend (Landing Outcome): Tracks first-stage landing success rate over time.

<https://github.com/FerAbrCode/Capstone/blob/main/EDAVisualization.ipynb>

EDA with SQL

- Queried total launches per year
- Calculated first-stage landing success rate by year
- Retrieved payload mass by orbit type
- Counted launches by orbit type
- Analyzed launch site usage over time
- Compared launch success by orbit type
- Queried payload mass vs. launch site
- Retrieved flight number vs. orbit type
- Analyzed landing success by booster reuse count
- Filtered failed landings with heavy payloads

<https://github.com/FerAbrCode/Capstone/blob/main/EDAsql.ipynb>

Build an Interactive Map with Folium

- **Markers** represent individual launch sites and infrastructure points (coastline, trainline, highway, city)
 - Used to pinpoint locations and provide popups with specific site or feature information
- **Marker Clustering** is used for areas with high launch activity, like Cape Canaveral
 - Prevents map clutter and helps reveal launch density patterns
- **Colored Markers** (Green/Red) differentiate successful (green) and unsuccessful (red) launch outcomes
 - Highlights performance trends at different locations
- **Lines (Polylines)** drawn between launch sites and infrastructure features
 - Visually connect elements to reflect logistical or geographical relationships (e.g., access to transportation, proximity to cities)
- **Distance Labels** display calculated distances between launch sites and nearby features (in km)
 - Makes proximity analysis immediately visible and interpretable

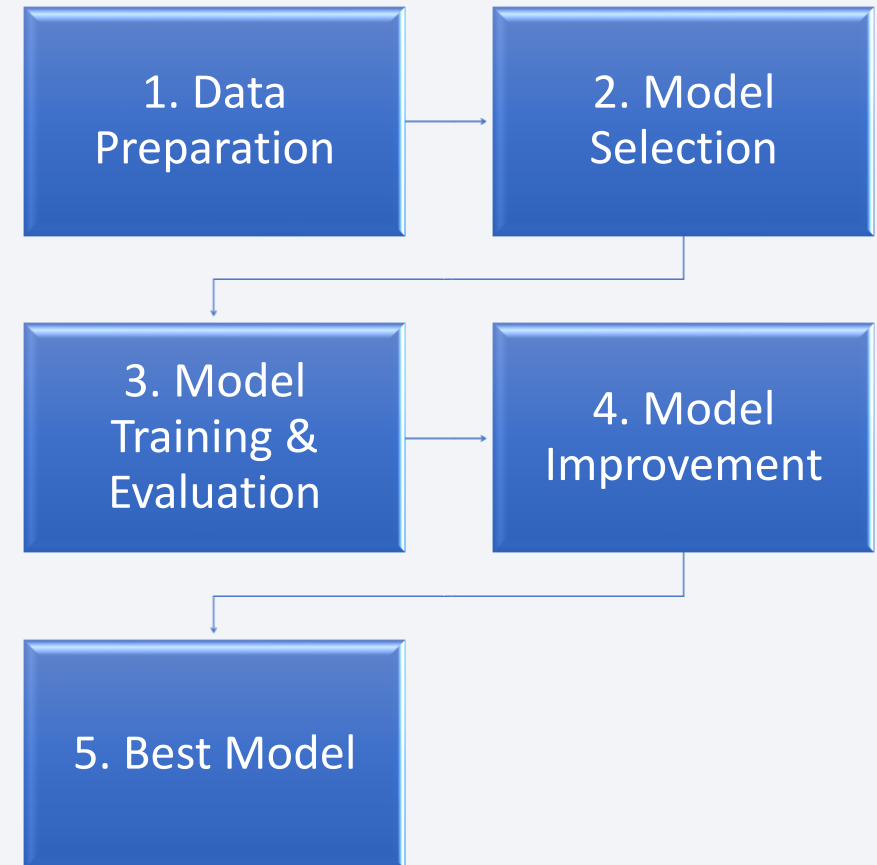
<https://github.com/FerAbrCode/Capstone/blob/main/Folium.ipynb>

Build a Dashboard with Plotly Dash

- **Pie Chart:** This chart shows the distribution of launch outcomes (success vs. failure) for a selected launch site.
- **Scatter Plot:** This plot displays the relationship between payload mass and launch outcome. Each point on the plot is colored by its booster version.
- **Dropdown Menu:** This menu lets you select a specific launch site or view data for all sites combined. It controls the data shown in both the pie chart and the scatter plot.
- **Range Slider:** This slider allows you to filter the data by selecting a specific range of payload mass. It controls the data shown in the scatter plot.
- <https://github.com/FerAbrCode/Capstone/blob/main/Dashboard.py>

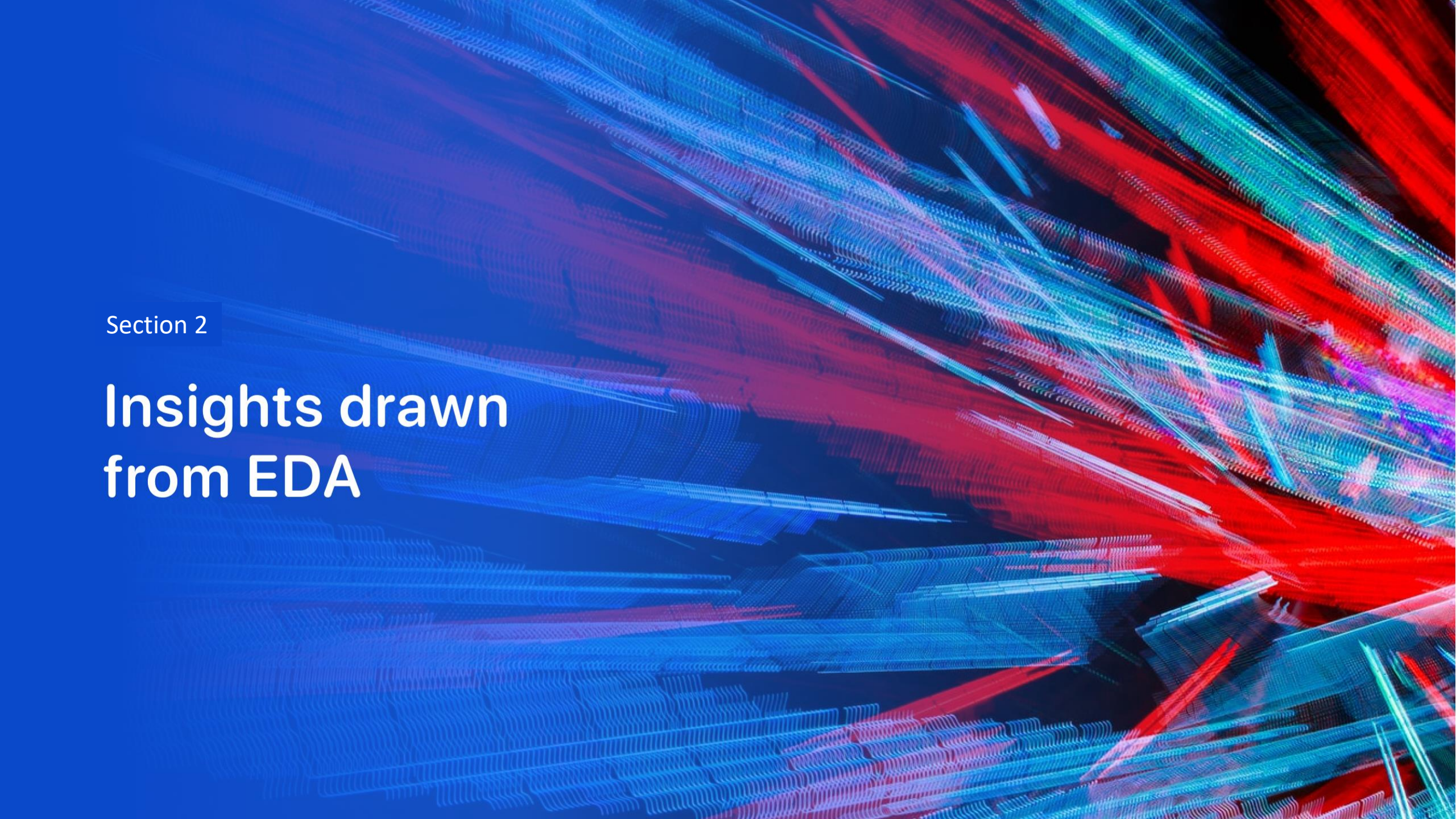
Predictive Analysis (Classification)

- **Data Preparation:** Standardized features and split the data into training and testing sets.
- **Model Selection:** Used Logistic Regression, SVM, and Decision Tree classifiers, K-Nearest Neighbors.
- **Hyperparameter Tuning:** Optimized each model using `GridSearchCV` with cross-validation.
- **Evaluation:** Evaluated all models on a test set, with all achieving the same accuracy score of 0.83333.
- **Conclusion:** All models performed almost equally well in predicting Falcon 9 first-stage landing success.
- <https://github.com/FerAbrCode/Capstone/blob/main/MachineLearning.ipynb>



Results

- **Exploratory Data Analysis Results:**
- **Launch Sites:** CCAFS SLC-40 has the highest success ratio (42.9%), with frequent launches compared to VAFB SLC-4E and KSC LC-39A.
- **Mission Outcomes:** 98 successes, 1 in-flight failure, reflecting high reliability.
- **Landing Outcomes:** Early attempts (2010-2017) included 10 no-attempts, 5 drone ship successes, and 5 failures; first ground pad success in December 2015.
- **Payload and Orbit:** No clear correlation between payload mass and success; GTO and VLEO missions faced higher risks due to complex requirements.
- **Booster Versions:** F9 B5 and B4 showed higher success rates than v1.0 and v1.1.
- **Interactive Analytics Demo:**
- **Folium Maps:** Visualized launch site locations (e.g., Cape Canaveral, Vandenberg) with color-coded markers (green for success, red for failure) and proximity analysis to infrastructure (e.g., 1-5 km to coastlines).
- **Plotly Dash:** Pie charts highlighted CCAFS SLC-40's success rate; scatter plots showed payload vs. outcome trends, with F9 B5 dominating successful heavy-payload missions.
- **Predictive Analysis Results:**
- Logistic regression achieved 83.3% accuracy in predicting launch success, outperforming DecisionTree models. Key predictors included booster version and launch site.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

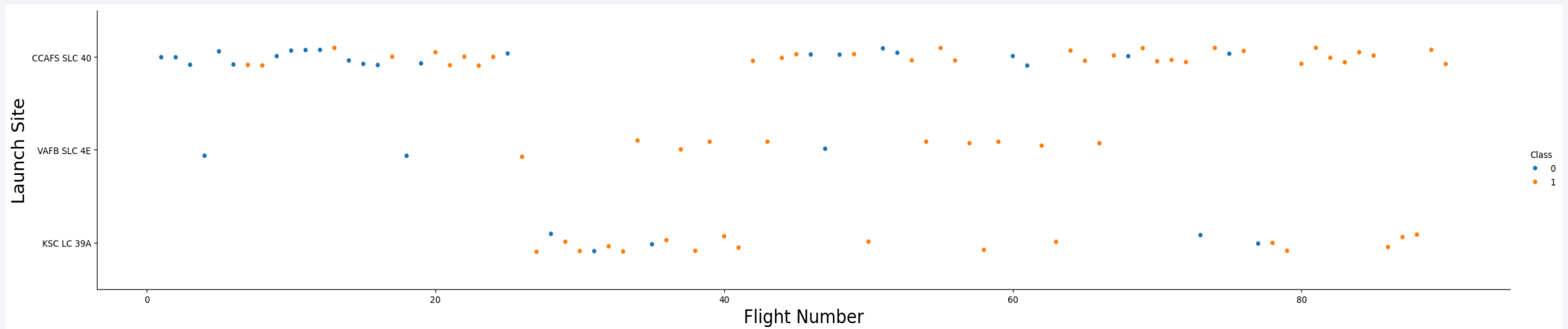
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Early Falcon 9 launches (2010–2015) were almost exclusively from **SLC-40**, leased in 2007
- After a **launch pad explosion** in 2016, SpaceX shifted many missions to LC-39A
- Post-2017, SLC-40 **resumed operations** and became the high-frequency pad, especially for Starlink launches. In 2023 alone, it hosted 50 launches
- **LC-39A** now handles crew missions, Falcon Heavy, and other **complex payloads** with higher prestige
- **VAFB SLC-4E** is used intermittently, depending on **orbital requirements** (e.g., polar or sun-synchronous orbits). It's essential for missions that can't launch from Florida

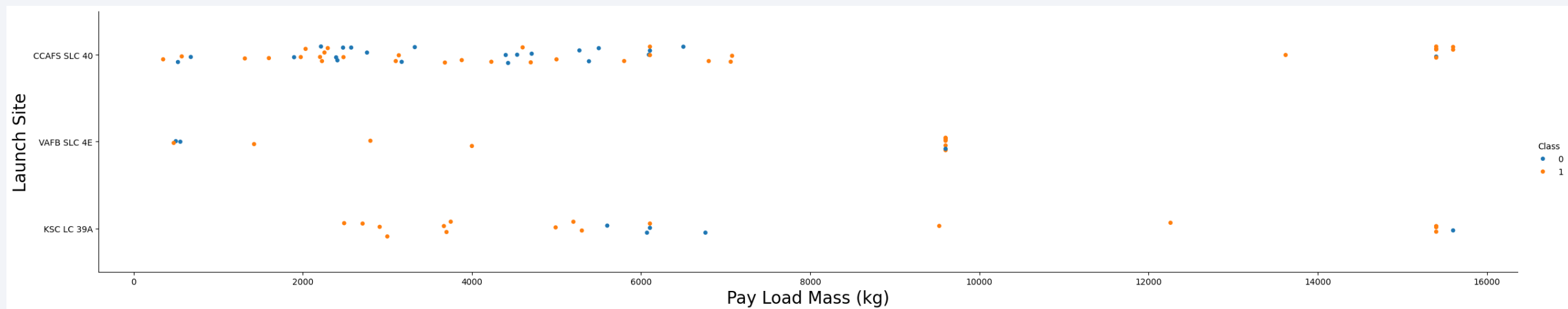
https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Launch_Complex_40



Payload vs. Launch Site

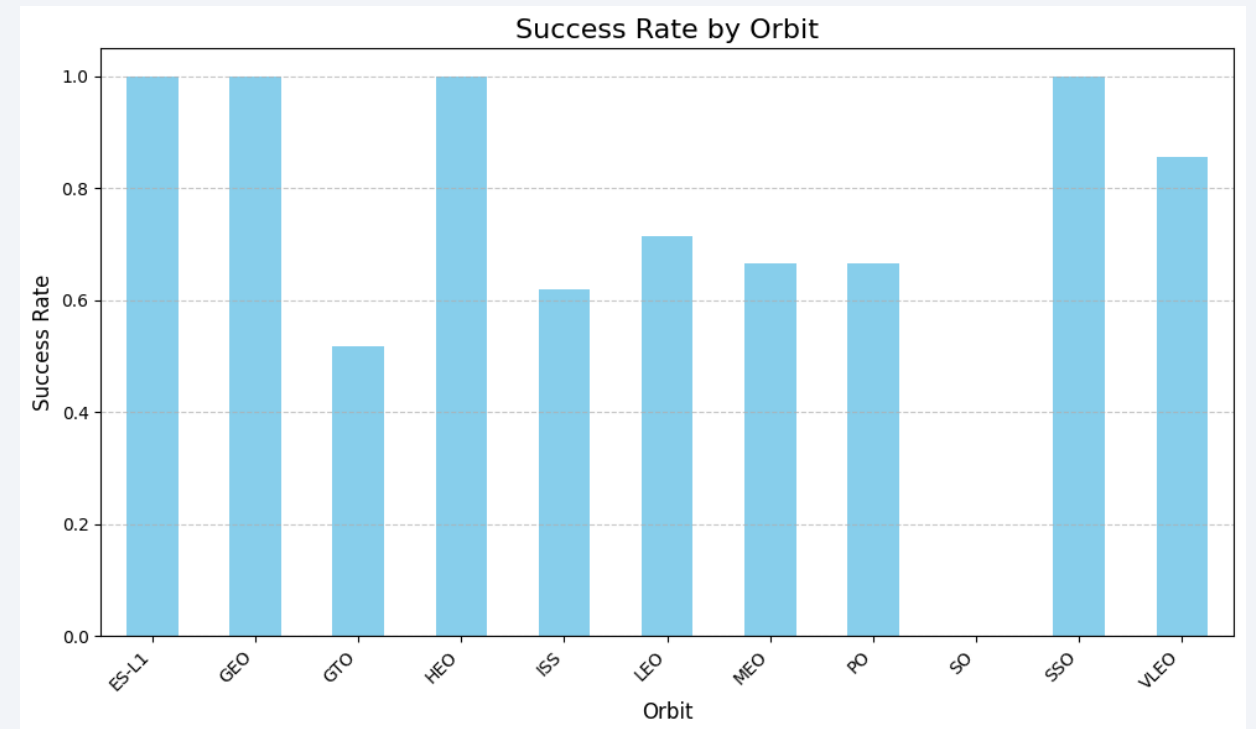
Efficiency Considerations

- **SLC-40** is optimized for **rapid, frequent launches**—perfect for Starlink and mid-weight payloads.
- **LC-39A** is reserved for **prestige and complexity**, including **Falcon Heavy** and **crewed missions**, which often push the upper payload limits.
- **VAFB SLC-4E** fills a niche role, enabling SpaceX to serve **government and commercial clients** needing specific orbital paths.



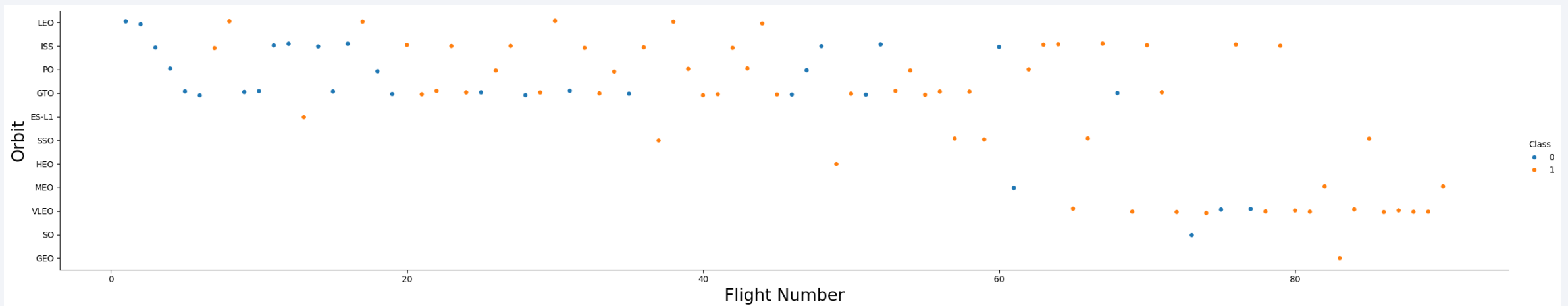
Success Rate vs. Orbit Type

- Early Falcon 9 missions targeted GTO and VLEO, when reliability was still maturing.
- GTO launches require complex burns and precise timing—more room for error.
- VLEO missions face intense atmospheric drag and tighter operational windows.
- Polar orbits (PO) often involve launches from VAFB, which had fewer launches and more experimental payloads.



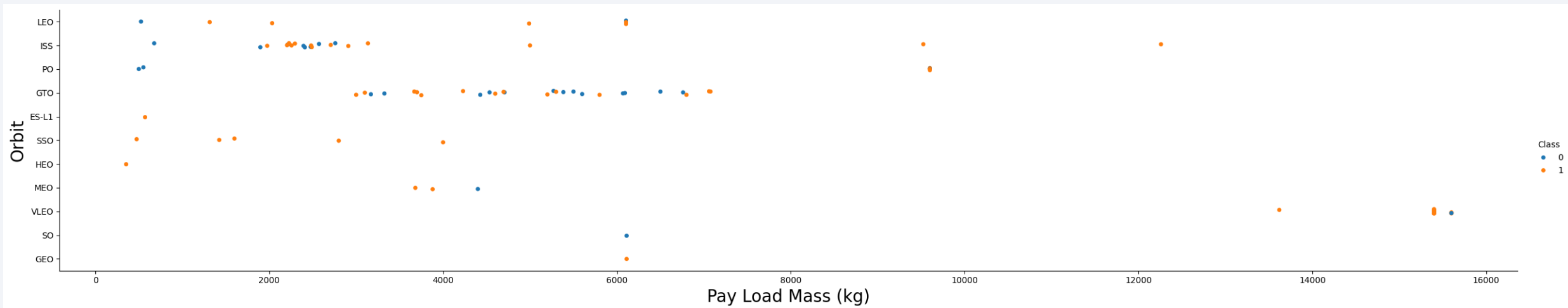
Flight Number vs. Orbit Type

- Market demand leads the payload and orbit type
- Technological maturity: Reliability and precision improved over time.
- Why Falcon 9 sometimes is expended (not recovered):
 1. **Heavy Payloads:** limited fuel margin for landing maneuvers
 2. **High-Energy Orbits** (e.g., GTO, Mars) that require more velocity.
 3. **Booster Reuse Limits:** If a booster is nearing the end of its usable life
 4. **Trajectory Constraints:** Certain launch profiles (e.g., polar or sun-synchronous from Florida) require unusual trajectories that may not allow for safe drone ship positioning



Payload vs. Orbit Type

- GTO missions often carry commercial satellites with standardized mass.
- ISS missions are constrained by NASA safety and docking requirements, leading to consistent payload sizes.
- VLEO missions are often experimental or tech demos, leading to:
- Occasional dense or high-mass instruments for close-range Earth observation



Launch Success Yearly Trend

Early Experiments (2013–2015)

- Initial attempts focused on **controlled descent** into the ocean.
- First successful landing: **Orbcomm OG2 mission** in Dec 2015 at LZ-1.

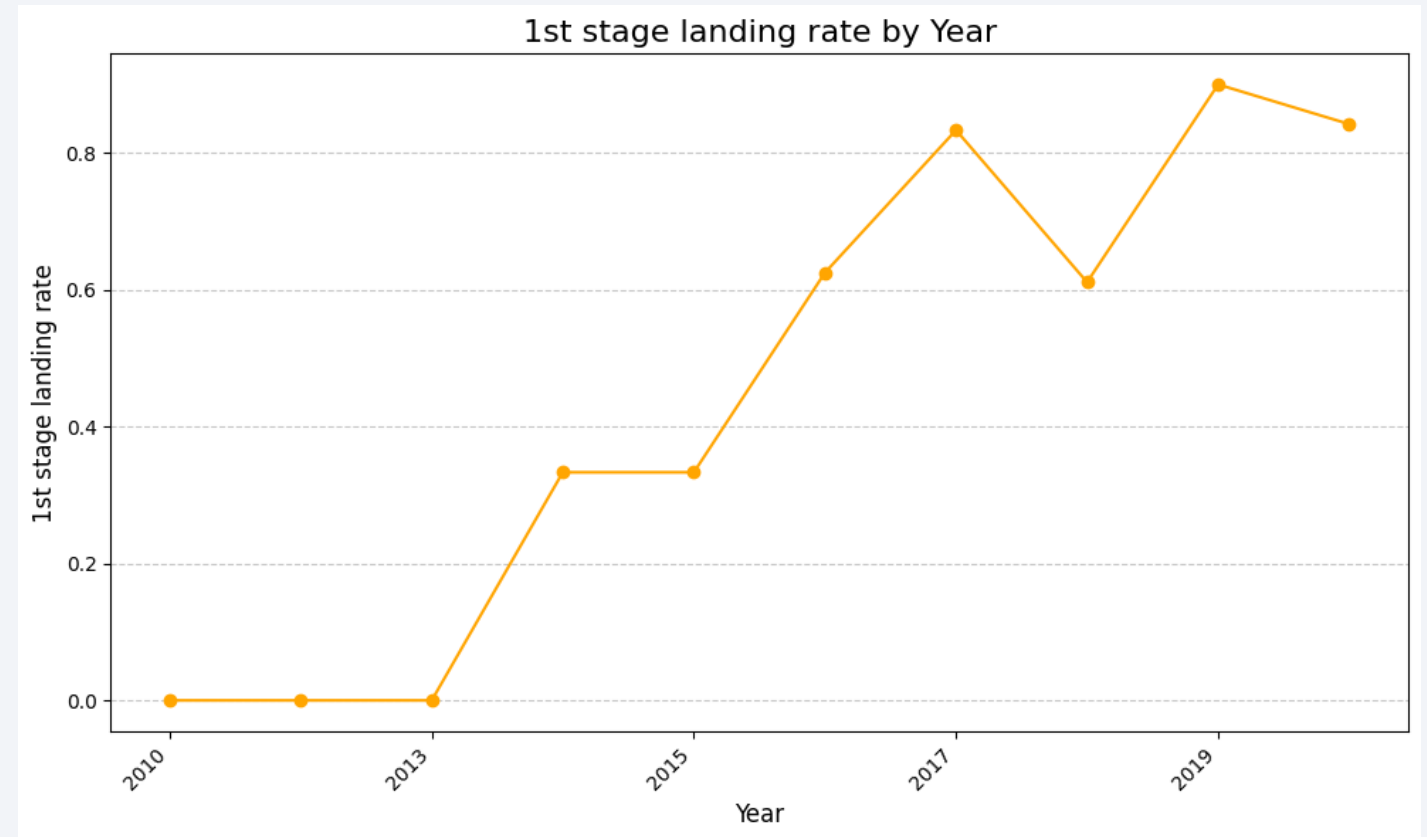
Technical Breakthroughs (2016–2018)

- Drone ship landings enabled recovery from high-energy missions.
- Block 5 introduced in 2018: designed for **10+ reuses** with minimal refurbishment.

Operational Maturity (2019–Present)

- Landings became routine, with boosters reused up to **29 times**.
- SpaceX now lands boosters on nearly every mission unless performance constraints prevent it.

https://en.wikipedia.org/wiki/Falcon_9_first-stage_landing_tests



All Launch Site Names

The SQL query retrieves unique launch site names from the SPACEXTABLE dataset. The result lists all distinct SpaceX launch sites, such as Cape Canaveral, Vandenberg, and Kennedy Space Center, used for Falcon 9 launches. This helps identify the variety of launch locations without duplicates, providing insight into SpaceX's operational geography.

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

The query retrieves the first five records from the SPACEXTABLE dataset where the launch site starts with "CCA" (Cape Canaveral). The result includes columns like Date, Time, Booster_Version, Launch_Site, Payload, Payload_Mass, Orbit, Customer, Mission_Outcome, and Landing_Outcome for launches from 2010 to 2013. It shows early Falcon 9 missions, primarily for NASA, with payloads like Dragon and CubeSats, and varying landing outcomes (e.g., parachute failures, no attempts). This highlights SpaceX's initial operations at Cape Canaveral.

```
SELECT * FROM SPACEXTABLE WHERE  
Launch_Site LIKE 'CCA%' LIMIT 5
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

The query is used to calculate the total payload mass (in kilograms) for all SpaceX Falcon 9 launches where the customer name includes "NASA". The result, 107,010 kg, represents the cumulative payload mass delivered to various orbits for NASA missions. This highlights SpaceX's significant contribution to NASA's space exploration efforts, including ISS resupply and other payloads.

```
SELECT SUM(PAYLOAD_MASS_KG_) AS TotalPayload FROM SPACEXTABLE  
WHERE Customer LIKE '%NASA%';`
```

TotalPayload

107010

Average Payload Mass by F9 v1.1

- The SQL query calculates the average payload mass (in kilograms) for each Falcon 9 v1.1 booster version in the SPACEXTABLE dataset. The results show the average payload mass for different F9 v1.1 boosters, ranging from 500 kg (B1003) to 4,707 kg (B1016). This highlights the variability in payload capacities across specific booster versions used in SpaceX's early missions.

```
SELECT Booster_Version,  
AVG(PAYLOAD_MASS__KG_) AS AveragePayload  
FROM SPACEXTABLE WHERE Booster_Version  
LIKE '%F9 v1.1%' GROUP BY Booster_Version
```

Booster_Version	AveragePayload
F9 v1.1	2928.4
F9 v1.1 B1003	500.0
F9 v1.1 B1010	2216.0
F9 v1.1 B1011	4428.0
F9 v1.1 B1012	2395.0
F9 v1.1 B1013	570.0
F9 v1.1 B1014	4159.0
F9 v1.1 B1015	1898.0
F9 v1.1 B1016	4707.0
F9 v1.1 B1017	553.0
F9 v1.1 B1018	1952.0

First Successful Ground Landing Date

The earliest date of a Falcon 9 launch with a landing outcome involving a ground pad is retrieved. The result, December 22, 2015, indicates the first successful ground pad landing in SpaceX's history, marking a key milestone in their reusable rocket program.

```
SELECT Date FROM SPACEXTABLE WHERE Landing_Outcome LIKE '%Ground  
pad%' ORDER BY Date LIMIT 1;
```

Date
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

The displayed query fetches the booster versions of Falcon 9 launches with a drone ship landing and payload masses between 4,000 and 6,000 kg. The results list specific F9 FT boosters (e.g., B1020, B1022, B1026, B1021.2, B1031.2) that meet these criteria, highlighting SpaceX's successful drone ship landings for mid-range payload missions.

```
SELECT Booster_Version FROM SPACEXTABLE WHERE  
Landing_Outcome LIKE '%Drone ship%' AND  
PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ <  
6000;
```

Booster_Version

F9 FT B1020

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

The SQL query counts the occurrences of each mission outcome in the SPACEXTABLE dataset. The results show 98 successful missions, 1 in-flight failure, 1 success with unclear payload status, and 1 additional success (possibly a formatting inconsistency). This summarizes SpaceX's Falcon 9 mission success rate, highlighting a high success rate with rare exceptions.

```
SELECT Mission_Outcome, COUNT(Mission_Outcome)
AS Total FROM SPACEXTABLE GROUP BY
Mission_Outcome;
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

The following query obtains the booster versions of Falcon 9 launches carrying the heaviest payload in the SPACEXTABLE dataset. The results list multiple F9 B5 boosters (e.g., B1048.4, B1049.4, B1051.3) that carried the maximum payload mass, indicating these boosters were used for SpaceX's heaviest missions, likely involving high-capacity payloads like Starlink satellites.

```
SELECT Booster_Version FROM SPACEXTABLE WHERE  
PAYLOAD_MASS__KG_ = (SELECT  
MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

The SQL query extracts the month, landing outcome, booster version, and launch site for Falcon 9 launches in 2015 with drone ship landing attempts. The results show three failed or precluded drone ship landings (January, April, June) using F9 v1.1 boosters at Cape Canaveral (CCAFS LC-40), reflecting SpaceX's early challenges with drone ship recovery in 2015.

```
SELECT strftime('%m', Date) AS Month, Landing_Outcome, Booster_Version,  
Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome LIKE '%Drone ship%' AND  
strftime('%Y', Date) = '2015';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
06	Precluded (drone ship)	F9 v1.1 B1018	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

The query counts the occurrences of each landing outcome for Falcon 9 launches between June 4, 2010, and March 20, 2017, sorted by frequency. The results reflect SpaceX's early landing attempts, with a mix of successes and challenges in developing reusable rocket technology.

```
SELECT Landing_Outcome,  
COUNT(Landing_Outcome) AS Total FROM  
SPACEXTABLE WHERE Date BETWEEN '2010-  
06-04' AND '2017-03-20' GROUP BY  
Landing_Outcome ORDER BY Total DESC
```

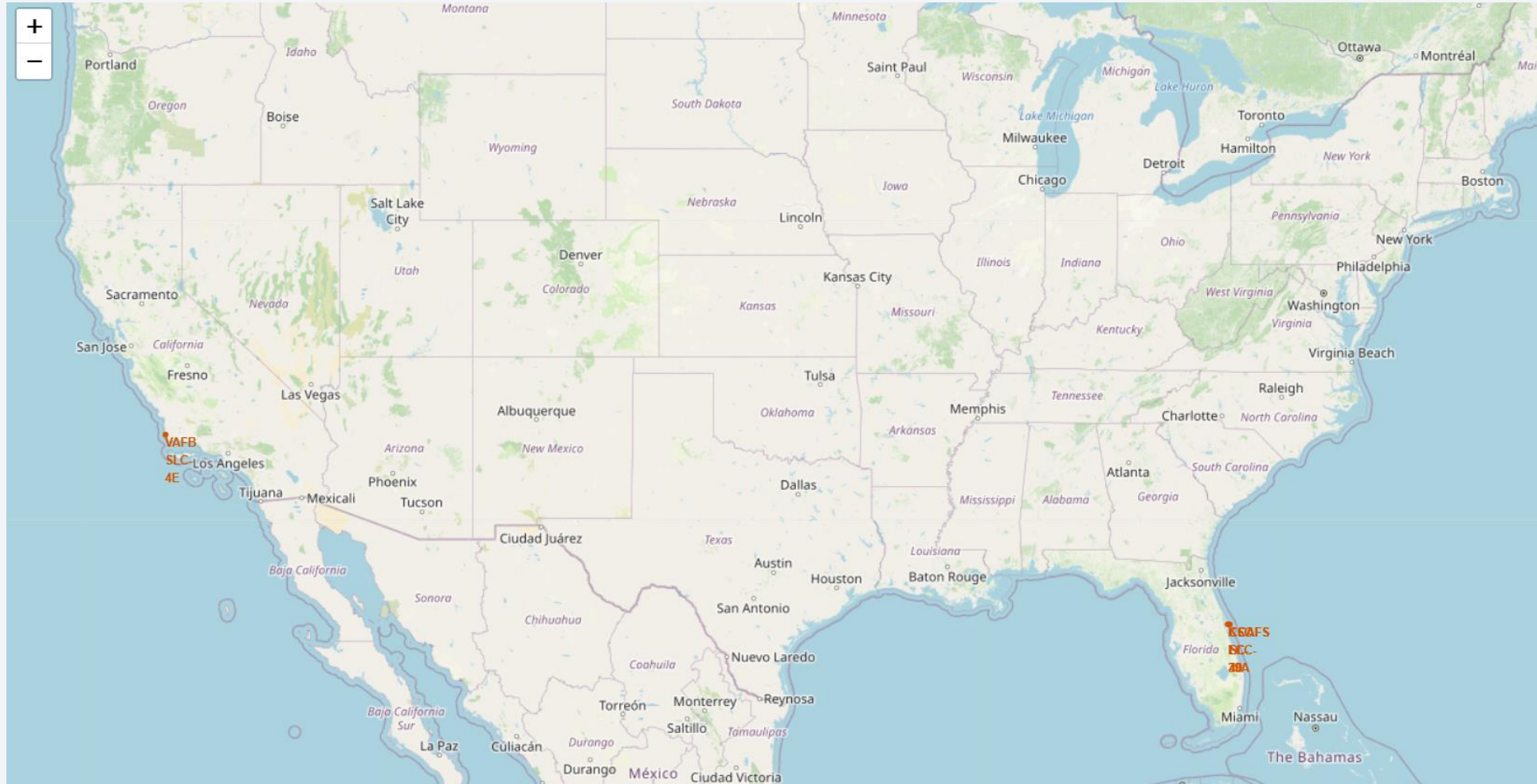
Landing_Outcome	Total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and the glowing lights of cities and continents against the dark background of space. The Earth's surface is a mix of blue oceans and dark landmasses, with numerous bright yellow and white lights indicating urban areas and infrastructure.

Section 3

Launch Sites Proximities Analysis

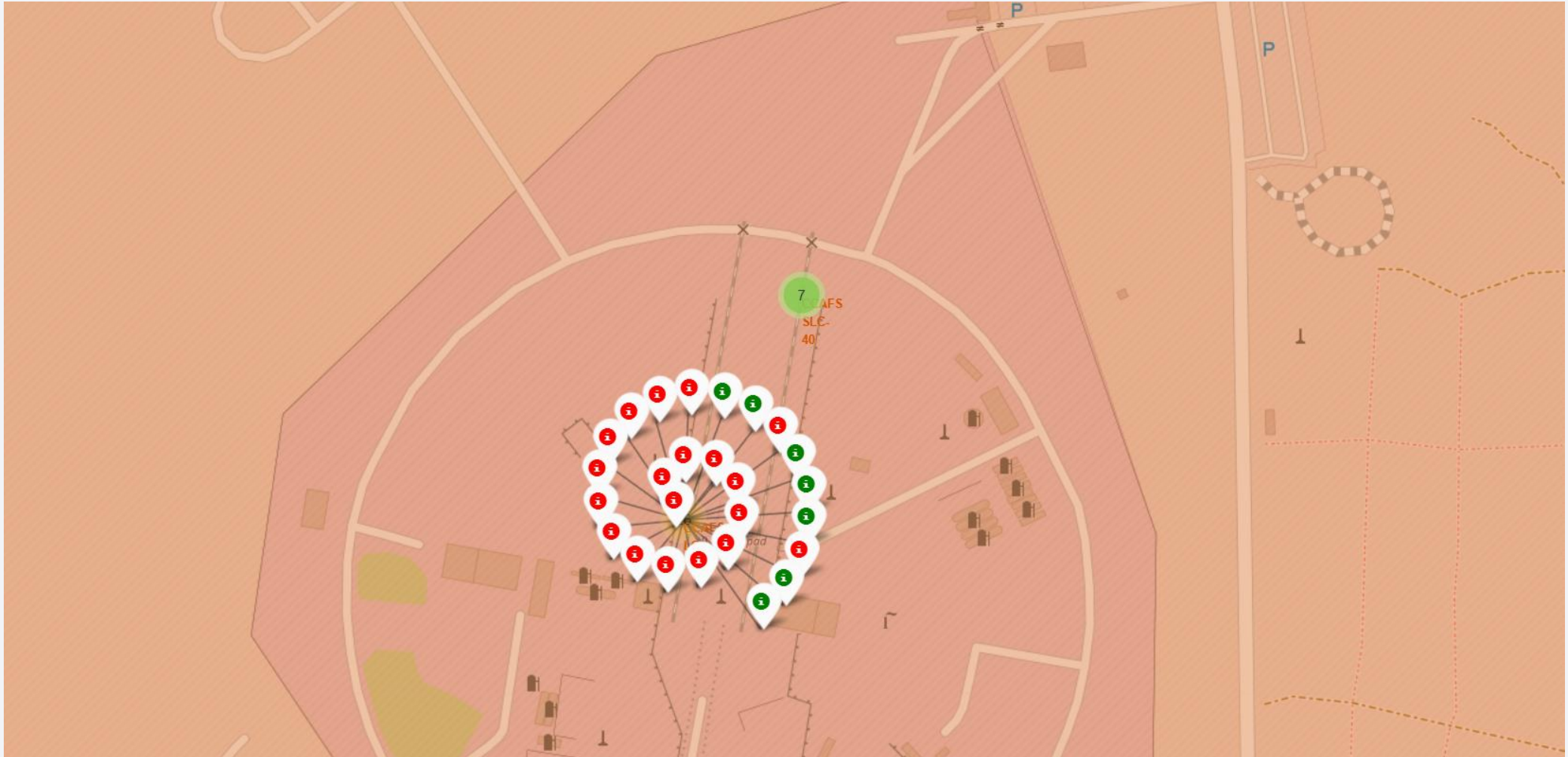
SpaceX's primary launch sites



SpaceX's primary launch sites

- The map shows SpaceX's primary launch sites, such as **Cape Canaveral** (Florida), **Vandenberg** (California), and **Kennedy Space Center** (Florida), reflecting their operational geography.
- The use of circles and markers emphasizes the spatial distribution of launch facilities, with East Coast sites (e.g., CCAFS, KSC) clustered closer together compared to the West Coast's Vandenberg site.
- The interactive popups enhance usability, allowing users to explore site-specific details, which is useful for analyzing launch frequency or site-specific success rates in the broader EDA context.

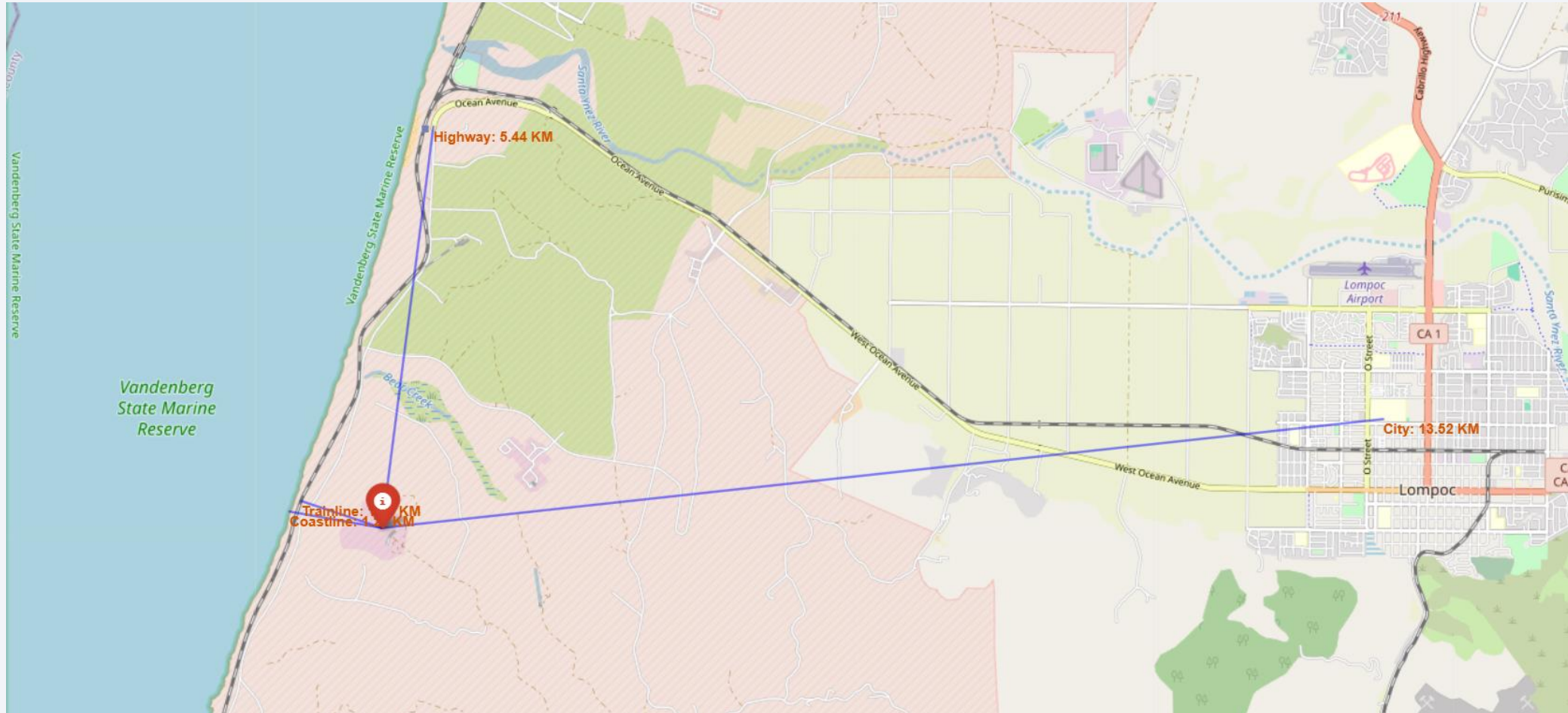
Interactive popups



Interactive launch site popups

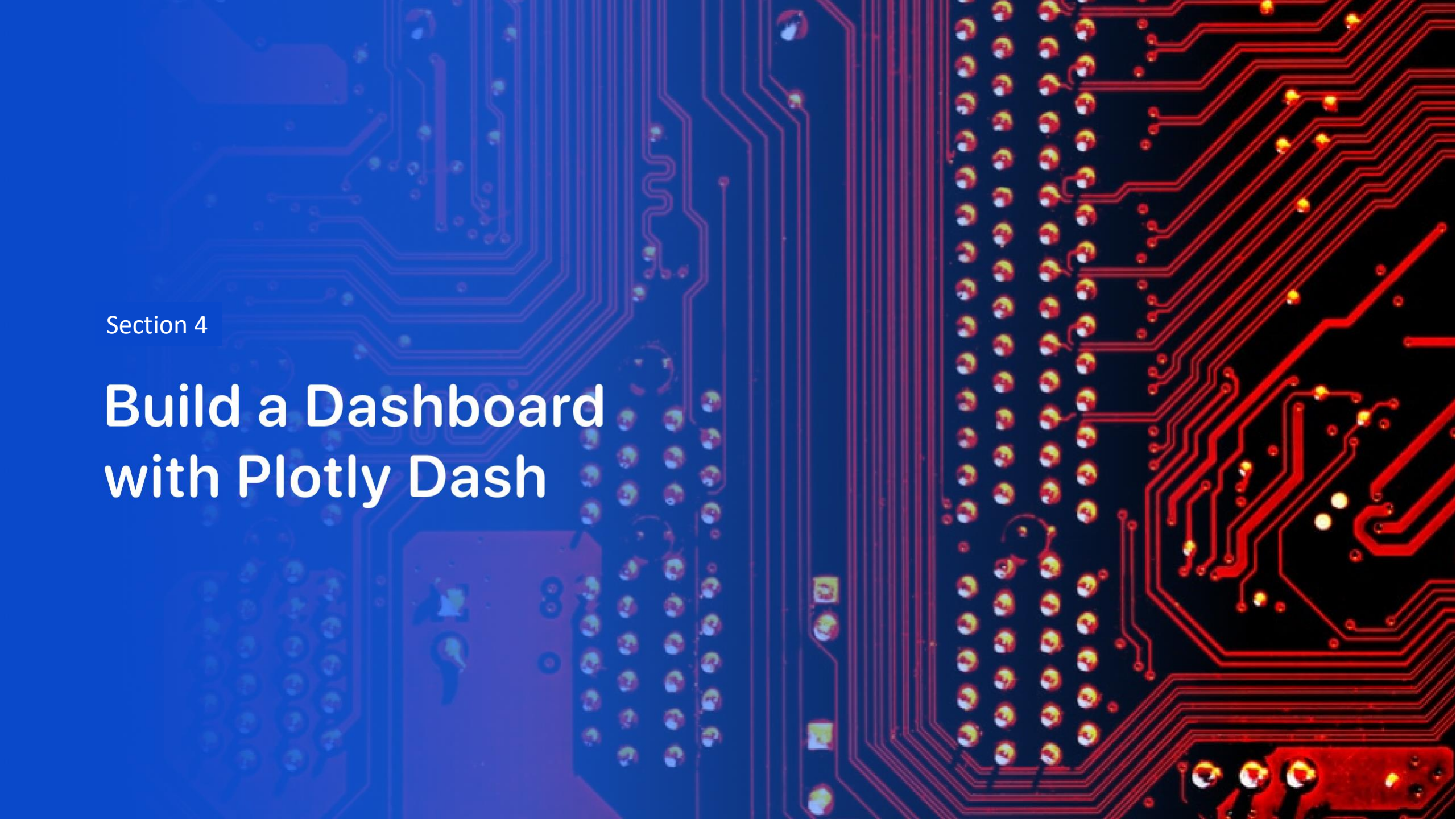
- The map visually distinguishes successful (green) and unsuccessful (red) launches across SpaceX's launch sites (e.g., CCAFS LC-40, KSC LC-39A, VAFB SLC-4E), highlighting the geographic distribution of launch outcomes
- Clustering reveals high launch activity at specific sites, with Cape Canaveral showing dense clusters due to frequent launches, while Vandenberg has fewer markers
- The interactive popups and clustering enhance user exploration, enabling analysis of site-specific success patterns, such as higher success rates at CCAFS compared to early Vandenberg launches

Proximity Analysis of Launch Site to Key Infrastructure



Proximity Analysis of Launch Site to Key Infrastructure

- The map displays the launch site (red marker) and its distance to four key infrastructure types: **Coastline**, **Trainline**, **Highway**, and **City**.
- Orange **distance labels** show the calculated distances using the Haversine formula (in kilometers), blue lines visually connect the launch site to each feature for spatial clarity.
- The coastline and trainline are closest ($\sim 1\text{--}1.5$ km), critical for safety and logistics.
- The highway offers convenient road access (~ 5.44 km).
- The nearest city (~ 13.52 km) provides essential support services and workforce access.

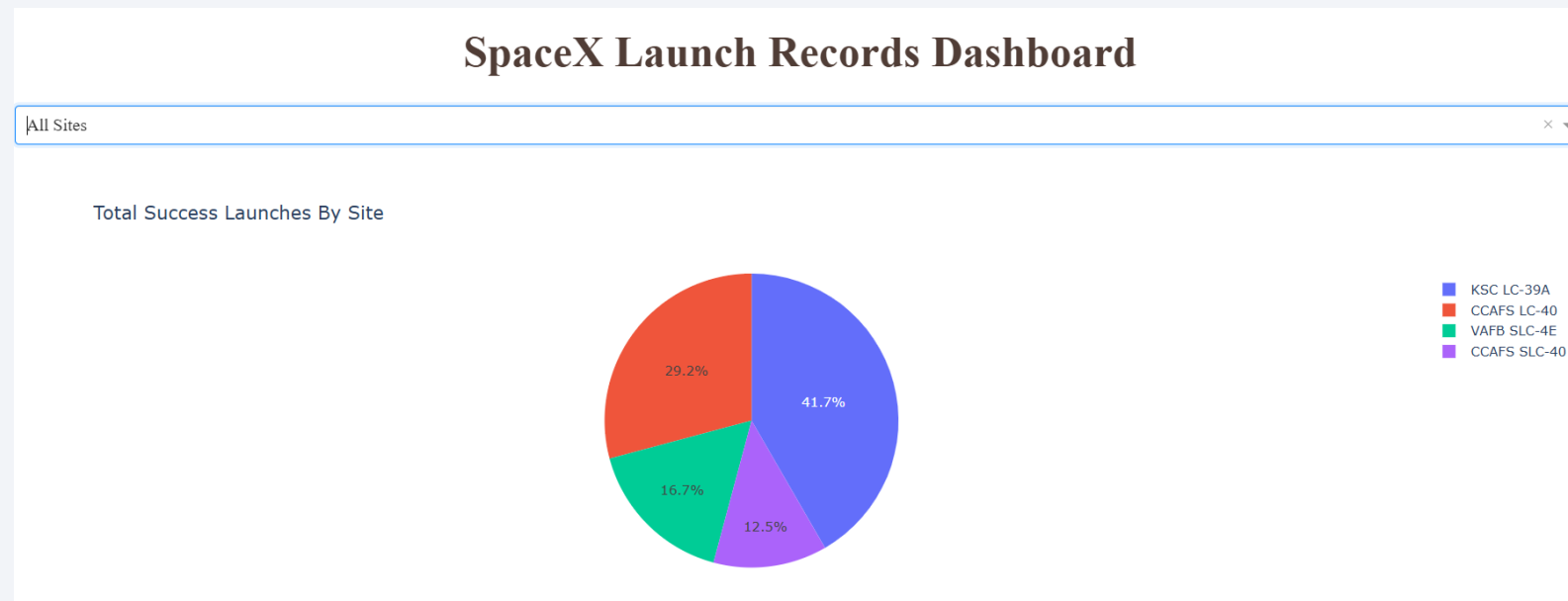


Section 4

Build a Dashboard with Plotly Dash

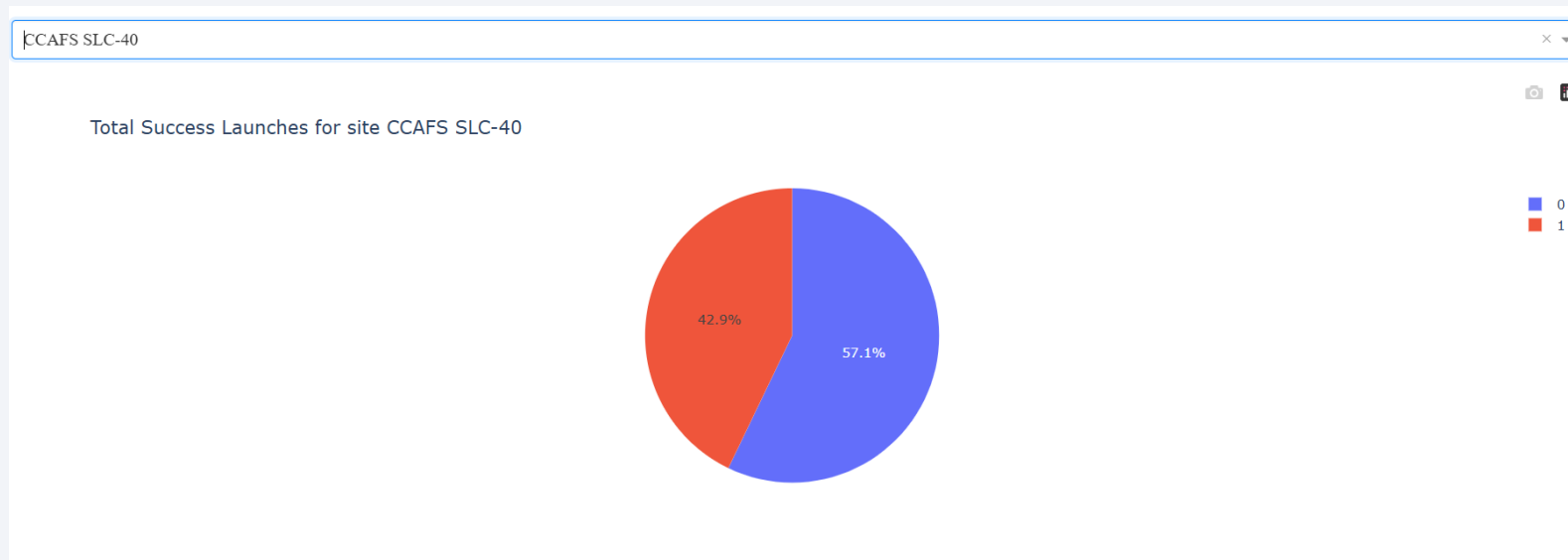
Launch Success Count for All Sites

This pie chart shows the overall success rate for all launch sites combined. The chart is divided into four sections for the launch sites: **CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40** representing **successful launches (class 1)**. The site KSC LC-39A has the largest portion of the pie with 10 successful launches.



Launch Success Ratio for CCAFS SLC-40

This pie chart focuses on the launch site with the highest success ratio, **CCAFS SLC-40**. The chart shows that this site has a **success rate of 42.9%**, which is a significant portion of its total launches. This finding is important for identifying the most reliable launch site in terms of mission success.

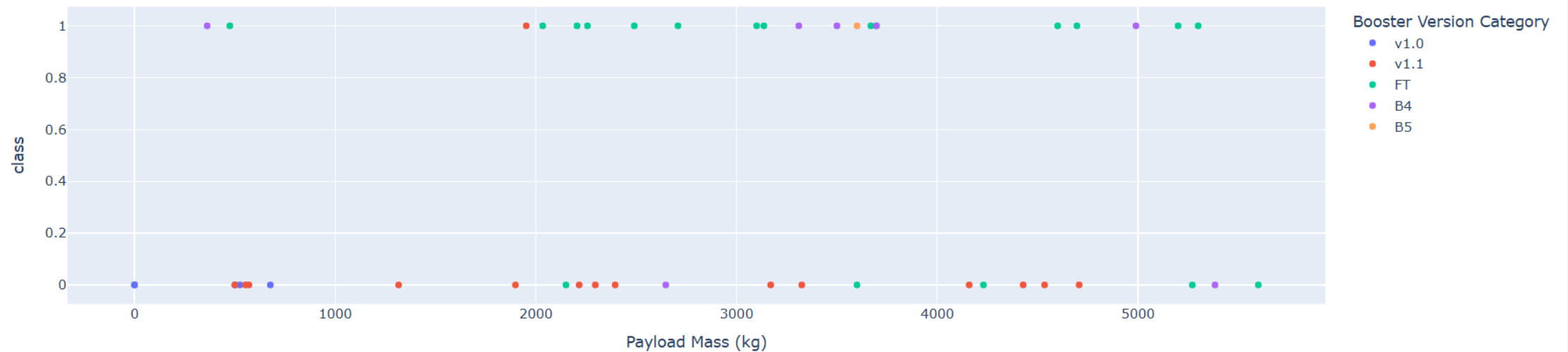


Payload vs. Launch Outcome Scatter Plot

Payload range (Kg):



Correlation between Payload and Success for all Sites



Payload vs. Launch Outcome Scatter Plot

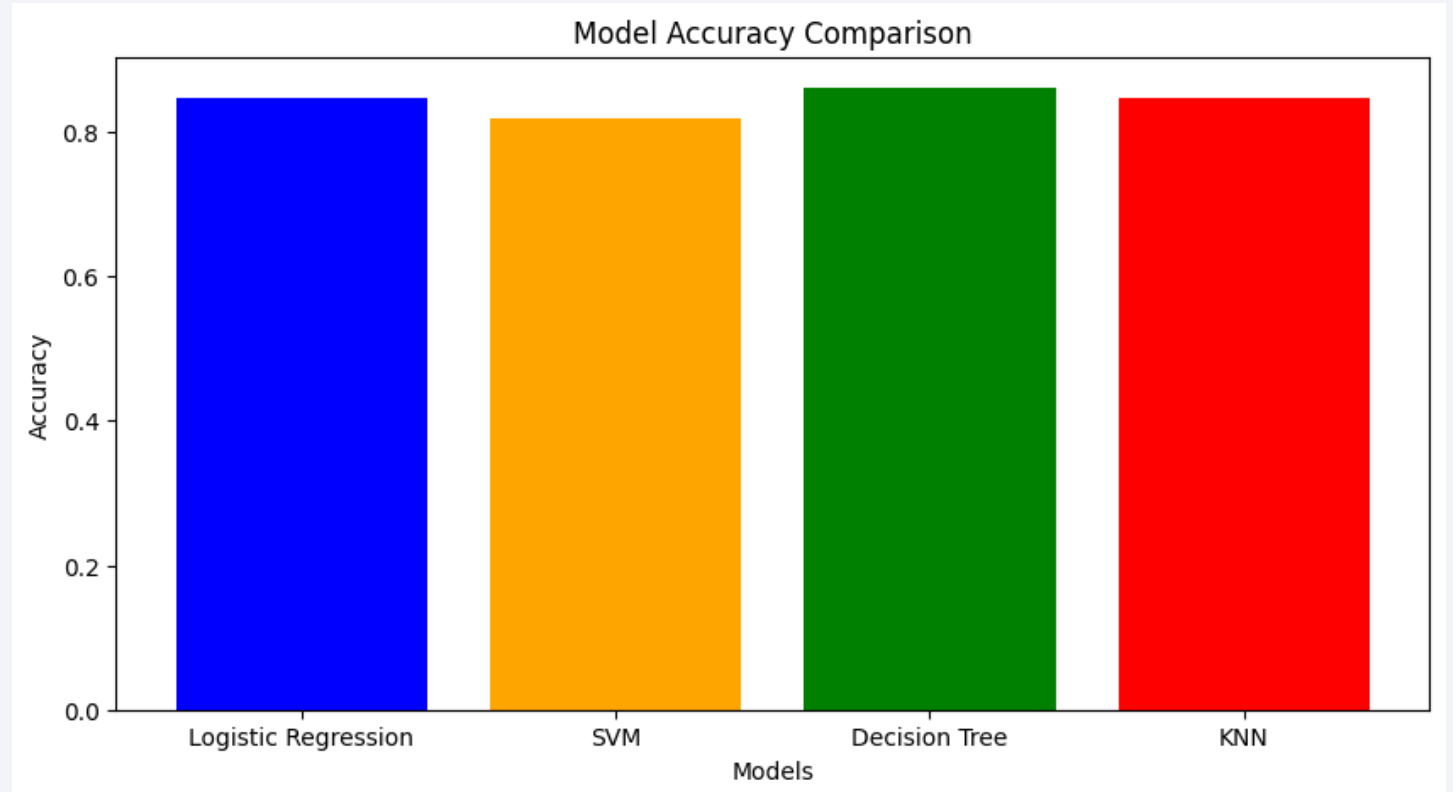
- The scatter plot visualizes the relationship between Payload Mass (kg) and launch outcome (class 0 for failure, 1 for success) for all sites. Each point is colored by the Booster Version Category. The plot reveals that:
- Booster Versions **v1.0** and **v1.1** appear to have the **lowest success rates**, as many of their points are in the failure (class 0) region.
- Booster Versions **FT**, **B5**, and **B4** are associated with the **highest success rates**, as their points are predominantly in the success (class 1) region.
- There doesn't appear to be a strong direct correlation between payload mass and success rate. Successes and failures are distributed across various payload ranges.

Section 5

Predictive Analysis (Classification)

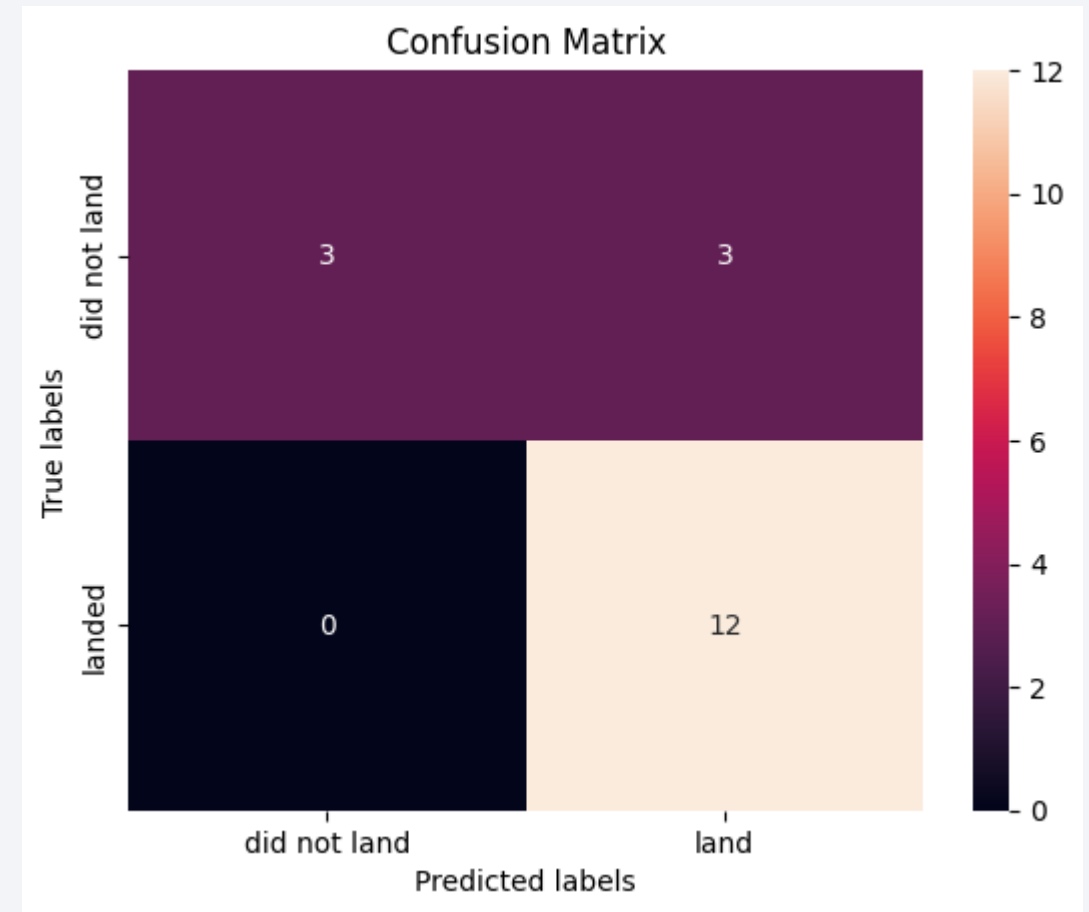
Classification Accuracy

- The best model is **LogisticRegression** with a score on test data **accuracy** of **0.8334**
- The rest of the models perform similarly, being the **DecisionTree** model the second-best



Confusion Matrix

- **Accuracy** = $(TP + TN) / \text{Total} = 83.3\%$
- The model is correct 83.3% of the time overall.
- **Precision** = $TP / (TP + FP) = 80\%$
- When the model predicts "landed," it's correct 80% of the time.
- **Recall** (Sensitivity) = $TP / (TP + FN) = 12 / (12 + 0) = 100\%$
- The model catches 100% of actual successful landings (no missed landings).
- **F1-Score** = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 88.9\%$
- Strong balance between precision and recall.



Conclusions

- **High Reliability:** SpaceX's Falcon 9 achieved a 98% mission success rate, with CCAFS SLC-40 as the most reliable launch site (42.9% success ratio), making it a preferred choice for future missions.
- **Landing Progress:** Early drone ship landings (2015) faced challenges, but ground pad success by December 2015 marked a turning point in reusable rocket technology.
- **Payload Flexibility:** Payload mass showed no strong correlation with launch success, indicating Falcon 9's versatility across diverse orbits (e.g., GTO, VLEO, ISS).
- **Predictive Power:** Logistic regression (83.3% accuracy) effectively predicts launch success, with booster version and launch site as key factors, supporting data-driven mission planning.

Appendix

- <https://github.com/FerAbrCode/Capstone>

Thank you!

